Aurora v1.2 Validation & Benchmark Suite Specification

AUR-VER-001 | Canonical Draft 1

1. Purpose

Define the tests and performance workloads required to validate the ISA, ABI, pipeline, streams, memory system, interrupts, compiler, simulator, and RTL.

2. Test Layers

- Level 0: instruction semantic unit tests
- Level 1: architectural-state and exception tests
- Level 2: pipeline hazard and pairing-window tests
- Level 3: memory, stream, DMA and interrupt integration tests
- Level 4: compiler and ABI conformance tests
- Level 5: benchmark kernels and end-to-end embedded workloads

3. ISA Conformance

- Every opcode/subfunction in the machine-readable ISA database
- All scalar, packed, L8/L16 and paired-64 modes
- All flag combinations and signed/unsigned corner cases
- Extension-word decoding and illegal-encoding traps
- Pair aliasing, odd-base faults and R12:R13 LR-liveness cases

4. Pipeline and Pairing Tests

- RAW/WAR/WAW across every relative window position
- One- and two-issue refill/compaction
- Branch misprediction with younger issued instructions
- Shared-resource conflicts for multiply, reduction, divide, LSU and streams
- Precise interrupt injection during every stage and multi-cycle phase

5. Benchmark Kernels

Category	Kernel	Primary metric	Architecture feature
DSP	FIR / IIR	cycles/sample	streams, MAC, circular addressing
DSP	FFT butterfly	cycles/butterfly	packed arithmetic, shuffles
DSP	INT8/INT16 matrix multiply	MACs/cycle	DOT, reductions, dual issue
Control	PID and Clarke/Park	cycles/control step	deterministic ALU/MAC

Memory	memcpy/memmove	bytes/cycle	streams, DMA, SRAM banks
Protocol	CRC / packet parse	bytes/cycle	L8, packed logic, shuffles
Crypto	AES / SHA-256	cycles/block	rotate, XOR, layout ops
RTOS	context switch	cycles/switch	ABI, stack, interrupts
System	DMA double buffer	sustained bandwidth	DMA, TCM, streams

6. Pass Criteria

- Bit-exact agreement among reference model, functional simulator and RTL
- Cycle-exact agreement among timing model and cycle-accurate RTL tests
- No unreviewed benchmark regressions above 3%
- 100% opcode and architectural-state coverage
- Formal proof of key safety properties where practical